

Bash Script -02

- Create a bash script to check if a directory is available or not.

```
#!/bin/bash
#adding direcorry file

if [ "-e /root/arun/scripts/kumar" ]

then

    echo "Directory exists"

else

    echo "Directory does not exist"
fi
```

```
5 b create.sh error.sh kumar response
[root@ip-172-31-0-121 scripts]# vi directory.bash
[root@ip-172-31-0-121 scripts]# chmod 755 directory.bash
[root@ip-172-31-0-121 scripts]# ./directory.bash
Directory exists
[root@ip-172-31-0-121 scripts]#
```

- Create a bash script to create multiple files.

```
#!/bin/bash
# mutliple files

for i in {1..5}
do
    touch "file$i.txt"
done

echo "Files created"
```

```
[root@ip-172-31-0-121 scripts]# vi multiple.bash
[root@ip-172-31-0-121 scripts]# chmod 755 multiple.bash
[root@ip-172-31-0-121 scripts]# ./multiple.bash
Files created
[root@ip-172-31-0-121 scripts]# ls
1 abc.bash case.sh dowhile.sh file5.txt linux seconds.sh within.sh
2 admin.bash commands.bash error.sh forloop.sh multiple.bash simple.sh
3 arun create.sh file1.txt helloworld.bash output.bash specify.sh
4 b define.bash file2.txt if.sh read users.sh
5 backup.sh definevariablees.bash file3.txt input.bash read.sh value.bash
a c directory.bash file4.txt kumar response.sh variables.bash
[root@ip-172-31-0-121 scripts]#
```

- Create a bash script to take a backup of a directory.

```
#!/bin/bash
#backup

NOW=$(date +"%a")
case $NOW in
    Mon)
        echo "Full backup";;
    Tue|Wed|Thu|Fri)
        echo "Partial backup";;
    Sat|Sun)
        echo "No backup";;
    *) ;;
esac
```

```
[root@ip-172-31-0-121 scripts]# backup.sh
bash: backup.sh: command not found
[root@ip-172-31-0-121 scripts]# vi backup.sh
[root@ip-172-31-0-121 scripts]# chmod 755 backup.sh
[root@ip-172-31-0-121 scripts]# ./backup.sh
Partial backup
[root@ip-172-31-0-121 scripts]#
```

- Create a bash script to install Nginx on an EC2 server.

```
[root@ip-172-31-0-121 scripts]# vi Nginx.bash
[root@ip-172-31-0-121 scripts]# chmod 755 Nginx.bash
[root@ip-172-31-0-121 scripts]# ./Nginx.bash
Last metadata expiration check: 0:54:20 ago on Fri Apr  3 08:56:12 2026.
=====
WARNING:
  A newer release of "Amazon Linux" is available.

  Available Versions:

  Version 2023.10.20260330:
    Run the following command to upgrade to 2023.10.20260330:

    dnf upgrade --releasever=2023.10.20260330

  Release notes:
    https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.10.20260330.html
=====
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:54:22 ago on Fri Apr  3 08:56:12 2026.
Package nginx-1:1.28.2-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
Nginx installed and running
[root@ip-172-31-0-121 scripts]#
```

```
#!/bin/bash
# install nginx

sudo yum update -y
sudo yum install nginx -y

sudo systemctl start nginx
sudo systemctl enable nginx

echo "Nginx installed and running"
~
```

- Create a bash script to install Apache Tomcat on an EC2 server.

```
#!/bin/bash
#Install apache tomcat
# Install Java
sudo yum install java-17-amazon-corretto -y

# Download Tomcat
cd /opt
sudo wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.85/bin/apache-tomcat-9.0.85.tar.gz

# Extract
sudo tar -xzf apache-tomcat-9.0.85.tar.gz
sudo mv apache-tomcat-9.0.85 tomcat

# Start Tomcat
sudo /opt/tomcat/bin/startup.sh

echo "Tomcat installed and started"
~
```

```
[root@ip-172-31-0-121 scripts]# vi Apache.bash
[root@ip-172-31-0-121 scripts]# chmod 755 Apache.bash
[root@ip-172-31-0-121 scripts]# ./Apache.bash
Last metadata expiration check: 1:01:02 ago on Fri Apr 3 08:56:12 2026.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
java-17-amazon-corretto                x86_64            1:17.0.18+9-1.amzn2023.1  amazonlinux      219 k
Installing dependencies:
glibc                                   x86_64            5.2.1-9.amzn2023.0.2    amazonlinux       48 k
libICE                                  x86_64            1.1.1-3.amzn2023.0.1    amazonlinux       76 k
libSM                                   x86_64            1.2.4-3.amzn2023.0.1    amazonlinux       45 k
libXi                                   x86_64            1.8.2-1.amzn2023.0.1    amazonlinux       42 k
libXinerama                             x86_64            1.1.5-6.amzn2023.0.1    amazonlinux       16 k
libXrandr                               x86_64            1.5.4-3.amzn2023.0.1    amazonlinux       29 k
libXt                                    x86_64            1.3.0-3.amzn2023.0.1    amazonlinux      183 k
libXtst                                 x86_64            1.2.5-1.amzn2023.0.1    amazonlinux       22 k
=====
Transaction Summary

```

```
Installed:
glibc-5.2.1-9.amzn2023.0.2.x86_64          java-17-amazon-corretto-1:17.0.18+9-1.amzn2023.1.
libICE-1.1.1-3.amzn2023.0.1.x86_64        libSM-1.2.4-3.amzn2023.0.1.x86_64
libXi-1.8.2-1.amzn2023.0.1.x86_64         libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64     libXt-1.3.0-3.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.1.x86_64

Complete!
--2026-04-03 09:57:15-- https://downloads.apache.org/tomcat/tomcat-9/v9.0.85/bin/apache-tomcat-9.0.8
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f9:3a:
Connecting to downloads.apache.org (downloads.apache.org)[135.181.214.104]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-04-03 09:57:16 ERROR 404: Not Found.

tar (child): apache-tomcat-9.0.85.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
mv: cannot stat 'apache-tomcat-9.0.85': No such file or directory
sudo: /opt/tomcat/bin/startup.sh: command not found
Tomcat installed and started
```

- Create a bash script to check if the Nginx service is running, if not running then script should start the service.

```
#!/bin/bash
#nginx start

if systemctl status nginx; then
    echo "Nginx is running"
else
    echo "Nginx not running. Starting..."
    systemctl start nginx
fi
```

```
[root@ip-172-31-0-121 scripts]# vi nginx.bash
[root@ip-172-31-0-121 scripts]# chmod 755 nginx.bash
[root@ip-172-31-0-121 scripts]# ./nginx.bash
Nginx is running
[root@ip-172-31-0-121 scripts]# vi nginx.bash
[root@ip-172-31-0-121 scripts]# ./nginx.bash
• nginx.service - The nginx HTTP and reverse proxy server
  Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
  Active: active (running) since Fri 2026-04-03 08:56:04 UTC; 1h 12min ago
  Main PID: 2166 (nginx)
  Tasks: 3 (limit: 1043)
  Memory: 4.3M
  CPU: 83ms
  CGroup: /system.slice/nginx.service
          └─2166 "nginx: master process /usr/sbin/nginx"
            └─2167 "nginx: worker process"
              └─2168 "nginx: worker process"
```

- Create a bash script for a calculator.

```
#!/bin/bash
#create calculator

echo "Enter first number:"
read a

echo "Enter second number:"
read b

echo "Choose operation (+ - * /):"
read op

case $op in
    +) echo "Result: $((a + b))" ;;
    -) echo "Result: $((a - b))" ;;
    \*) echo "Result: $((a * b))" ;;
    /) echo "Result: $((a / b))" ;;
    *) echo "Invalid operation" ;;
esac
```

```
[root@ip-172-31-0-121 scripts]# vi calculator.bash
[root@ip-172-31-0-121 scripts]# chmod 755 calculator.bash
[root@ip-172-31-0-121 scripts]# ./calculator.bash
Enter first number:
1
Enter second number:
2
Choose operation (+ - * /):
+
Result: 3
[root@ip-172-31-0-121 scripts]#
```

- Create a bash script to check if a directory exists, if not then create a directory.

```
#!/bin/bash
# new directory or create new one

DIR="/root/arun/scripts/sai"

if [ ! -d "$DIR" ]; then
    mkdir -p "$DIR"
    echo "Directory created"
else
    echo "Directory already exists"
fi
```

```
[root@ip-172-31-0-121 scripts]# vi newdir.bash
[root@ip-172-31-0-121 scripts]# chmod 755 newdir.bash
[root@ip-172-31-0-121 scripts]# ./newdir.bash
Directory created
```

- Create a bash script to delete the last 3 lines of a file.

```
#!/bin/bash

FILE="test.txt"

head -n -3 "$FILE" > temp.txt && mv temp.txt "$FILE"

echo "Last 3 lines deleted"
```

```
[root@ip-172-31-0-121 scripts]# touch test.txt
[root@ip-172-31-0-121 scripts]# vi delete.bash
[root@ip-172-31-0-121 scripts]# chmod 755 delete.bash
[root@ip-172-31-0-121 scripts]# ./delete.bash
Last 3 lines deleted
[root@ip-172-31-0-121 scripts]# ls
1      abc.bash      commands.bash      file1.txt          input.bash         read.sh            variables.bash
2      admin.bash     create.sh          file2.txt          kumar              response.sh        within.sh
3      arun           define.bash        file3.txt          linux              seconds.sh
4      b              definevariablees.bash file4.txt          multiple.bash      simple.sh
5      backup.sh     delete.bash        file5.txt          newdir.bash        specify.sh
Apache.bash c              directory.bash     forloop.sh         nginx.bash         test.txt
Nginx.bash calculator.bash dowhile.sh         helloworld.bash    output.bash        users.sh
a      case.sh        error.sh           if.sh              read               value.bash
[root@ip-172-31-0-121 scripts]#
```

- Bash script to monitor cpu and if it is more than 80% then send email notification.

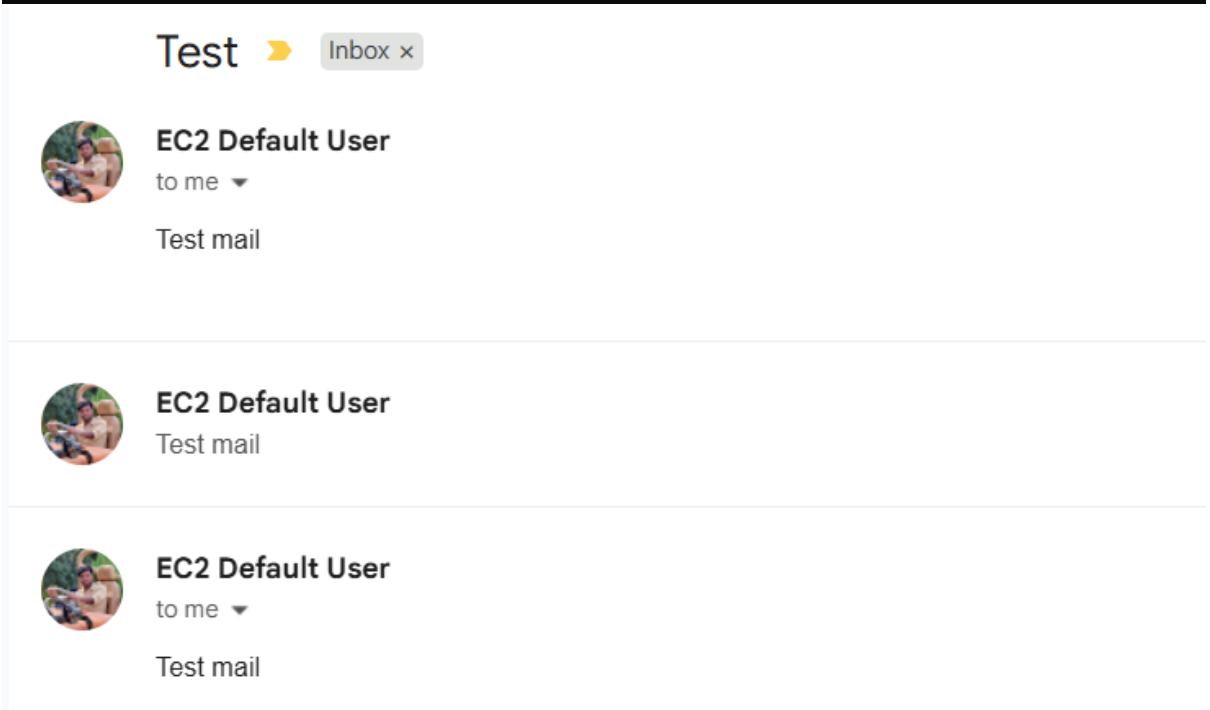
```
#!/bin/bash

CPU=$(top -bn1 | grep "Cpu(s)" | awk '{print 100 - $8}' | cut -d. -f1)

if [ "$CPU" -gt 80 ]; then
    echo "High CPU usage: $CPU%" | mail -s "CPU Alert" arunsai.mailme@gmail.com
fi
```

```
[root@ip-172-31-0-121 ec2-user]# cd /arun/scripts
[root@ip-172-31-0-121 scripts]# vi monitor_cpu.bash
[root@ip-172-31-0-121 scripts]# ./monitor_cpu.bash

[root@ip-172-31-0-121 scripts]# sudo systemctl restart postfix
[root@ip-172-31-0-121 scripts]# echo "Test mail" | mail -s "Test" arunsai.mailme@gmail.com
[root@ip-172-31-0-121 scripts]# ./monitor_cpu.bash
Current CPU: 3%
```



- Bash script to monitor disk space and if it is more than 80% then send email notification.

```
[root@ip-172-31-0-121 scripts]# ./disk.bash
Current Disk Usage: 35%
Disk usage is normal
[root@ip-172-31-0-121 scripts]# vi disk.bash
[root@ip-172-31-0-121 scripts]# chmod 755 disk.bash
[root@ip-172-31-0-121 scripts]# ./disk.bash
Current Disk Usage: 35%
Alert sent
[root@ip-172-31-0-121 scripts]#
```

```
#!/bin/bash

THRESHOLD=30
EMAIL="arunsai.mailme@gmail.com"

# Get disk usage of root (/)
USAGE=$(df / | awk 'NR==2 {print $5}' | tr -d '%')

echo "Current Disk Usage: $USAGE%"

if [ "$USAGE" -gt "$THRESHOLD" ]; then
    MESSAGE="$(date) - Disk usage is high: $USAGE%"

    # Send email (may fail without SMTP setup)
    echo "$MESSAGE" | mail -s "Disk Alert" $EMAIL

    echo "Alert sent"
else
    echo "Disk usage is normal"
fi
```

Disk Alert Inbox x



EC2 Default User

to me ▼

Sat Apr 4 07:14:56 UTC 2026 - Disk usage is high: 35%

- Bash script to monitor memory and if it is more than 80% then send email notification.

```
#!/bin/bash

THRESHOLD=30
EMAIL="arunsai.mailme@gmail.com"

# Get memory usage percentage
MEM_USAGE=$(free | awk '/Mem/ {printf("%.0f"), $3/$2 * 100}')

echo "Current Memory Usage: $MEM_USAGE%"

if [ "$MEM_USAGE" -gt "$THRESHOLD" ]; then
    MESSAGE="$(date) - Memory usage is high: $MEM_USAGE%"

    # Send email (works only if SMTP configured)
    echo "$MESSAGE" | mail -s "Memory Alert" $EMAIL

    echo "Alert sent"
else
    echo "Memory usage is normal"
fi
```

```
[root@ip-172-31-0-121 scripts]# vi memory.bash
[root@ip-172-31-0-121 scripts]# vi memory.bash
[root@ip-172-31-0-121 scripts]# chmod 755 memory.bash
[root@ip-172-31-0-121 scripts]# ./memory.bash
Current Memory Usage: 70%
Alert sent
[root@ip-172-31-0-121 scripts]#
```

Memory Alert ➡ Inbox x



EC2 Default User

to me ▼

Sat Apr 4 07:21:01 UTC 2026 - Memory usage is high: 70%

Exercise: Crontab Entries

April 5th Midnight

5th of Every November, January, June if it is a Thursday

1. At 05 and 27th minutes of 9, 10, 11 hours every day

34 minutes of 9th hour on 15th August

Every midnight

1. Every Weekend (Saturday night 11:59 PM)
2. After every reboot

```
#!/bin/bash
# April 5th at midnight
0 0 5 4 * /root/nginx_bash

# Midnight on the 5th of November, January, June ONLY if it's Thursday
0 0 5 1,6,11 4 /root/nginx_bash

# At 05 and 27 minutes of 9, 10, 11 every day
5,27 9-11 * * * /root/nginx_bash

# At 34 minutes of 9th hour on 15th August
34 9 15 8 * /root/nginx_bash

# Every midnight (daily)
0 0 * * * /root/nginx_bash

# Every weekend (Saturday night at 11:59 PM)
59 23 * * 6 /root/nginx_bash

# After every reboot
@reboot /root/nginx_bash
```

```
[root@ip-172-31-0-121 ~]# crontab -e
crontab: installing new crontab
[root@ip-172-31-0-121 ~]# crontab -l
#!/bin/bash
# April 5th at midnight
0 0 5 4 * /root/nginx_bash

# Midnight on the 5th of November, January, June ONLY if it's Thursday
0 0 5 1,6,11 4 /root/nginx_bash

# At 05 and 27 minutes of 9, 10, 11 every day
5,27 9-11 * * * /root/nginx_bash

# At 34 minutes of 9th hour on 15th August
34 9 15 8 * /root/nginx_bash

# Every midnight (daily)
0 0 * * * /root/nginx_bash

# Every weekend (Saturday night at 11:59 PM)
59 23 * * 6 /root/nginx_bash

# After every reboot
@reboot /root/nginx_bash
[root@ip-172-31-0-121 ~]#
```

